Week 4 - Arrays: Lecture Solutions

Software Development II (4COSC010C) · 20 February 2024 · typed up from projector photos

Solutions shown on the projector in the Week 4 (arrays) class, 20 February 2024. Comments are the lecturer's.

Swap two elements in a String array

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String[] array = {"Alex", "Max", "Charlie", "Bob", "Ada", "Jim"};

        System.out.println("Enter 2 indices to swap between 0 and 5: ");
        int index1 = scanner.nextInt();
        int index2 = scanner.nextInt();

        //swap values
        String temp = array[index1];
        array[index1] = array[index2];
        array[index2] = temp;

        System.out.println("Array after swapping: ");
        for (String element : array) {
            System.out.print(element + ", ");
        }
    }
}
```

q3 - Swap with index checking

```
import java.util.Scanner;

public class q3 {
    public static void main(String[] args) {
        double[] values = {34.5, 55.5, 67.4, 675.34, 43645.87, 45.45, 456.66, 67.55};
        Scanner input = new Scanner(System.in);
        System.out.println("Enter 2 values: ");
        int i = input.nextInt();
        int j = input.nextInt();

        try {
            double temp = values[i];
            values[i] = values[j];
            values[j] = temp;

            System.out.println("Array after swapping:\n");
            for (double elements : values) {
                System.out.print(elements + ", ");
            }
        } catch (Exception e) {
            System.out.println("Index out of bounds");
        }
    }
}
```

q4_a - Even numbers in set A

```
public class q4_a {
    public static void main(String[] args) {
        int[] A = {2, 3, 4, 5, 6, 7, 8, 9};
        System.out.println("Even numbers in set A:");
        for (int num : A) {
            if (num % 2 == 0)
                System.out.println(num);
        }
    }
}
```

q4_b - Numbers whose double is also in the array

```
public class q4_b {
    public static void main(String[] args) {
        int[] array = {2,3,4,5,6,7,8,9};           //declared the array A
        int[] chosenNumbers = new int[10];          //created a new array of size 10. ty
        int j = 0;                                  //count to keep track of the chosen
        for (int num1 : array) {                    //enhanced for loop to loop through
            for (int num2 : array) {                //loop again to check if the result
                if (num2 == num1*2) {               //checking if there are values in A
                    chosenNumbers[j] = num1;        //adding it to the new array
                    j++;                            //incrementing the index
                }
            }
        }
        for (int element : chosenNumbers) {         //printing the values identified.
            System.out.println(element);
        }
    }
}
```

q5 - 2D array: marks for 5 students x 3 subjects

```
import java.util.Scanner;

public class q5 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);     //created a new scanner instance
        double[][] marks = new double[5][3];          //created a new 2D array.

        for (int i = 0; i < 5; i++) {                 //for loop to iterate between the
            System.out.println("Enter marks for Student " + (i + 1) + ":");
            for (int j = 0; j < 3; j++) {             //for loop between the subjects fo
                System.out.print("Enter mark for Subject " + (j + 1) + ": ");
                marks[i][j] = scanner.nextDouble();   //storing the user inputs into the
            }
        }

        System.out.println("\nAverage marks for each student:");
        for (int i = 0; i < 5; i++) {
            double sum = 0;
            for (int j = 0; j < 3; j++) {
                sum += marks[i][j];
            }
            double average = sum / 3;
            System.out.println("Student " + (i + 1) + ": " + average);
        }
```

```
    }
}
```

q6 - Binary search

```
public class q6 {
    public static void main(String[] args) {
        int[] runs = {5, 12, 28, 34, 47, 52, 65, 71, 80, 99};
        int scoreToFind = 47;

        int matchNumber = binarySearch(runs, scoreToFind);

        if (matchNumber != -1) {
            System.out.println("Batsman scored " + scoreToFind + " runs in match " + (ma
        } else {
            System.out.println("Batsman did not score " + scoreToFind + " runs in any ma
        }
    }

    public static int binarySearch(int[] arr, int target) {
        int left = 0;
        int right = arr.length - 1;

        while (left <= right) {
            int mid = left + (right - left) / 2;

            if (arr[mid] == target) {
                return mid;
            } else if (arr[mid] < target) {
                left = mid + 1;
            } else {
                right = mid - 1;
            }
        }
        return -1; // Indicates that the target value was not found in the array
    }
}
```

The end of both println messages in q6 was cut off in the photo; the wording after "runs in" is a reconstruction.